Real-Time Spotify Lyric Translator + Song Vibe Board

Project Proposal: GAIT Fall 2025 Final Project
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Introduction:

Music is one of the most universal forms of expression, yet language barriers often limit a listener's ability to understand the true meaning of a song. Many listeners regularly enjoy music in languages they do not speak, often without understanding the emotional depth, cultural context, or lyrical significance embedded in the artist's message. Our project aims to address this gap by building a **Real-Time Spotify Lyric Translator + Song Vibe Board**, a tool that uses generative AI systems to simultaneously translate lyrics and visually represent the emotional "vibe" of a song.

The system integrates Spotify's API, OpenAI's generative models, and an AI image-generation tool to create an engaging experience where listeners get real-time translations and a dynamic, mood-matched visual background generated from the lyrical content. This promotes cultural understanding, encourages cross-language music exploration, and creates a more immersive experience.

Motivation:

As multilingual music lovers, we've experienced situations where we sing along to foreign-language songs without fully understanding their meaning. This can lead to misunderstandings or judgments, as in our example of singing along to "Si Antes Te Hubiera Conocido," unaware of some of its themes. Beyond personal embarrassment, this highlights a larger cultural issue: music crosses borders faster than understanding does.

Our motivation stems from:

- **Bridging linguistic and cultural gaps** by making global music instantly accessible.
- **Encouraging cultural appreciation and learning** through real-time translation.
- **Enhancing listener experience** by pairing music with AI-generated visual "vibe boards" that reflect emotional tone, themes, and energy.

This project creatively combines multiple GAITs to build a unique, interactive experience that aligns with the course's emphasis on generative AI applications.

Project Idea:

The Real-Time Spotify Lyric Translator + Song Vibe Board allows users to play a song in Spotify and receive:

- **Real-time lyric translation** into any target language (using OpenAI translation models).
- **A continuously updating "vibe board"**: a set of colors, shapes, or abstract patterns representing the emotional tone and themes extracted from the lyrics.
- **A synchronized UI panel** showing the original lyric line, its translated version, and a dynamically shifting background.

System Diagram / Process Flow

Spotify API → Lyric Fetching → GAIT #1 (OpenAI LLM Translation) → GAIT #2 (OpenAI Emotion/Vibe Classification) → GAIT #3 (Image Generator / Style Generator) → Front-End UI

Step-by-Step Flow

- **Spotify Player & API Integration**

    - Authenticate the user.
    - Retrieve the currently playing track and timed lyric segments (via Spotify API / Musixmatch integration).

- **Real-Time Lyric Translation (GAIT #1)**

    - Each lyric segment is sent to an OpenAI translation model (e.g., GPT-4.1 / GPT-4o).
    - The API returns a translation in the user's target language.

- **Vibe Extraction / Emotion Analysis (GAIT #2)**

    - The same lyric segment (either original or translated) is fed into an OpenAI model for:
      - Emotion recognition (e.g., melancholy, energetic)
      - Theme detection (love, heartbreak, motivation, nostalgia)
      - Suggested color palette or mood keywords

    - Output will be formatted into a prompt for the image generator.

- **AI Image / Color Board Generation (GAIT #3)**

- Using a generative image model (e.g., OpenAI Image Model, Midjourney, or Stable Diffusion Web API), the system generates:
  - Abstract visuals
  - Color gradients
  - “Aura board” representing the vibe

- This updates periodically or per verse/chorus.

- **Front-End Display**

  - UI shows:
    - Current lyric line (original + translated)
    -Background vibe board generated by AI
    - Smooth transitions between visual states